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Unwrapping the Customer Value concept and its application within an Industrial Manufacturing Firm

A Case Study within Husqvarna

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Abstract

Traditional manufacturing companies were reliant on selling their product centred offerings towards customers. However, technological developments, volatile market environments as well as calls for more personalised experiences from customers, have made the traditional product-centred approaches less effective. Manufacturing companies have thus been looking to adapt to the previously mentioned circumstances by transitioning towards a mix of products together with physical and digital services. Although there is a big interest from manufacturing companies to undergo such changes to adapt, no clear empirical evidence has emerged for cases where companies have undergone a full transition towards product-service offerings. Despite the lack of empirical evidence, companies are still interested in slowly building up towards offering comprehensive product-service offerings. One such company is the Swedish based manufacturing company, Husqvarna. A company which has a rich history in product development and has shown its capabilities for adaptation in the past through the manufacturing of different types of products, but also in the present with their successful releases of automatic lawn mowers. However, the company has further planned to expand upon their efforts of pursuing product-service offerings. Through internal discussions with the company, it was decided that this pursuit requires the key understanding of how the concept of customer value changes between pure product offerings and integrated

product-service offerings. It was thus decided that the purpose of this thesis report would be to look internally, to focus on understanding the different interpretations of the customer value concept as well as the practical implications of these interpretations.

The main methodology used in this thesis report was a case study approach, which included informants from three separate parts of the organisation. A total of twelve interviews were performed, where ten interviews were with employees of the company, that involved managers, product specialists and product owners. The last two interviews involved professors from KTH as theoretical guidance early in the research process. This data gathering involved literature search together with semi-structured interviews. Data was then analysed by using automated transcribing tools, the transcriptions were then coded through an inductive approach inspired by Gioia et al. (2012).

Findings from this study highlight that different teams within separate departments can have similar interpretations regarding the customer value concept. That being the belief that delivering experiences to customers is the key part of the offering that creates value for customers. The company is also more recently reliant on internal collaborations between different departments, communication issues were found in the cases where different departments did not have a similar interpretation of customer value. Thus, highlighting the importance of new communication strategies within organisations reliant on internal collaborations for more subjective concepts such as customer value. The concept of customer value was used practically in early pre-development stages, for gathering and understanding customer needs, but also in testing phases at the end of development for validating and verifying that the intended experience is delivered. The last thing to highlight is that both the case company and other companies need to develop or improve their relational capabilities both with customers and other network actors to offer better services. In the case of manufacturing companies, this means developing relationships beyond just business and creating opportunities and empowering customers to take part in more parts of the development process, not just the early and late phases.

KTH Industriell teknik och management

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Utforskning av begreppet Kundvärde och dess tillämpning inom ett traditionellt tillverkningsföretag

Robert Bunea

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Sammanfattning

Traditionella tillverkningsföretag har varit beroende av att sälja produkter till sina kunder. Däremot har framgångar i teknologisk utveckling tillsammans, mer varierande och osäkra marknadsförhållanden samt en önskan av mer personliga upplevelser från kunder har gjort att produktcentrerade sätt har blivit mindre effektiva. Således har tillverkningsföretag letat efter hur man ska anpassa till dessa omständigheter genom att övergå till att erbjuda en blandning av produkter tillsammans med fysiska och digitala tjänster. Det finns ett stort intresse från tillverkningsföretag att genomgå sådana förändringar, så saknas det tydlig empirisk data som visar att det finns företag som har lyckats med övergången. Trots att det saknas empirisk data, är företag fortfarande intresserade av att sakta röra sig till att erbjuda mer omfattande produkt-tjänst erbjudanden. Ett sådant företag är det svenskbaserade tillverkningsföretaget Husqvarna. Ett företag med en djup och rik historia inom produktutveckling och har visat sina förmågor för anpassning genom tiderna genom skapandet av varierande produkter, men också i nuläget med företagets robotgräsklippare. Företaget har däremot större planer på att fullfölja sina intressen i att erbjuda en blandning av produkter och tjänster. Interna diskussioner med företaget har lett till att syftet för denna rapport skulle vara att undersöka olika sätt att förklara och definiera kundvärde samt dess praktiska tillämpning.

Den huvudsakliga metodiken som har använts i denna rapport var baserat på en fallstudie, vilket inkluderade informanter från tre olika avdelningar inom organisationen. Där totalt tolv intervjuer har gjorts, där tio intervjuer har varit med managers, produktansvariga och produktspecialister. De två sista intervjuerna involverade professorer från KTH för vägledning i skapandet av rapportens teoretiska bakgrund tidigt i processen. I denna datainsamling innefattade både litteraturstudier och semi-strukturerade intervjuer. Den data som har samlats inte har sedan analyserats genom att använda ett automatiserad transkriberings verktyg, transkriberingarna har kodats genom en induktiv metod inspirerad från Gioia et al., (2012).

Resultaten från denna studie har visat att olika grupper inom separata avdelningar i företaget kan ha olika tolkningar och definitioner av begreppet kundvärde. Där tolkningarna påpekar att erbjudandet av upplevelser är en central del av ett erbjudande som är värdefullt för kunden. Företaget är också starkt beroende av interna samarbeten mellan olika avdelningar, kommunikationsproblem har hittats bland avdelningar som inte delade samma tolkningar eller definitioner av kundvärde begreppet. Detta visar hur viktigt det är att använda sig av nya kommunikationsstrategier inom en organisation som är starkt beroende av internt samarbete gällande subjektiva koncept såsom kundvärde. Kundvärdebegreppet används praktiskt först tidigt i produktutvecklingsprocessen, i formen av att förstå kundens behov, men också senare i processen för att testa erbjudanden till kunder och verifiera att den förväntade upplevelsen är levererad till kunden. Det sista intressanta resultatet berör både företaget i fallstudien men även andra företag, nämligen att ett större fokus måste läggas i att utveckla och förbättra sina förmågor i att skapa relationer med både kund och andra aktörer för att skapa och erbjuda bättre tjänster. Tillverkningsföretag måste därmed utveckla relationer endast för ekonomiska syften, och även fokusera på att skapa möjligheter som tillåter både aktörer och kunder för att vara med genom hela utvecklingsprocessen, inte i endast ett tidigt samt sent stadie.

Foreword

Firstly, I would like to personally thank Husqvarna for their active collaboration, support and giving the opportunity throughout this thesis report. Especially, the informants who took time to partake in the interviewing process with their valuable inputs and their honesty, together with the project supervisors, Gustaf Lagercrantz and Johan Simonsson.

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1. Introduction

1.1 Background

A traditional manufacturing company within Sweden wishes to expand their offerings beyond the current product-centred approach towards more integrated product-service offerings. The company in question is a product development pioneer, namely Husqvarna, that continues to operate to this day within areas such as forestry, gardening, and construction sectors all around the globe. integrated product-service offerings are seen as an opportunity by Husqvarna as it can lead to enabling new revenue streams, closer customer communication and increased customisability for this manufacturing company.

Despite the advantages to integrated product-service offerings, the company is in early stages of transitioning. Therefore, one of the areas that Husqvarna have been interested in gaining more insight into is understanding customer value when transitioning towards product-service offerings, as it is a core component of any offering. Compared to traditional offerings where the main selling point is the product and its features, product-service systems provide additional value to the customer through the services offered together with it through more personalised experiences (Kuijken, Gemser & Wijnberg, 2017). Lastly, the lack of successful transitions has made it tricky for researchers to analyse how some companies have succeeded in such a transition despite the large amount of theory available regarding multi-level benefits of product-service offerings (Brissaud et.al., 2022). A reason is that traditional manufacturing companies typically lack critical innovation capabilities to undergo such a transition, a few examples being a service-oriented culture, working with intangible artifacts together with closer department interactions (Wallin, Parida & Isaksson, 2015; Brissaud et.al., 2022). Despite the lack of successful examples of such transitions, it is still vital for companies to understand their own interpretations and application of the customer value concept. The concept of customer value has been researched for decades, both from a business perspective on how to better leverage it and from the customer perspective to better

understand how customers make decisions and value offerings (Zeithaml et al., 2020). Additionally, It is a line of research which is still divided despite the advancements because of the subjective nature of the concept and the introduction of new paradigms, such as the service-dominant logic and an ever-changing environment which introduces new opportunities together with challenges. This thesis work will therefore hope to expand upon the theory of customer value by exploring its practical implementations throughout development projects.

1.2 Purpose and definition

The goal of this master thesis paper is to investigate how different departments interpret and apply the concept of customer value within the same industrial manufacturing company. Specifically focusing on the application of the concept when developing new product-service offerings compared to previously only developing product offering. This will require an understanding of the organisation's current development process and key activities throughout the process for different parts of the organisation. Thus, leading to the two main research questions listed below.

RQ 1: What are the interpretations regarding the concept of customer value within different teams throughout a development project?

RQ 2: How is the concept of customer value used within the new product development process?

2. Theoretical background

This section will introduce and explore the key theoretical concepts that will create the foundation to understand customer value and its practical implications in the development of novel offerings. Firstly, by investigating customer value as a concept, its different interpretations and evolution over time from a research perspective. Followed by investigating how different organisational factors and processes affect an organisation's likelihood to develop innovative offerings.

2.1 Modern Development of Customer Value

Value is a central concept within the business industry, where understanding how customers think of value can provide a vital competitive advantage for any firm (Christensen, 2010; Woodruff, 1997). The research regarding customer value is predominantly popular within the industry of marketing (Zeithaml et al., 2020). The role of marketing in an organisation is vital and concerned with acquiring information and knowledge on customer needs and other competitive offerings, inform and orient decisions at all levels of the organisation towards benefiting the customer and aid the organisation in developing intriguing value propositions based on customer needs (Webster, 1992). Over time the research around customer value developed, resulting in many different emerging perspectives. Understanding the basis upon which these different perspectives are built upon will be vital in understanding definitions and views regarding customer value in an organisational context. Thus, the two most prominent perspectives are presented and analysed, the classical economic perspective and the new paradigm of service dominant logic will be introduced and their definitions and basis for the perspectives will be analysed.

2.1.1 Classical Economic Perspective of Customer Value

Studies of customer value in the past decades originated from a mix of classical psychology and economic theories which formed the basis for future studies (Gakkarza et.al, 2011). During this period, research was mainly focusing on the relationship between price and quality, where it was argued and believed that price is positively related to

quality. Based on the fact that both are being governed by supply and demand across different industries, where prices will find an accepted level in accordance with the quality available on the market (Scitovszky 1944; Stiglitz 1987). Despite this being a purely economic perspective, it was still a base on which early value perception research was built upon. Initially, studies took a single-cue approach, meaning that only the relationship between price and quality was studied. Although this relationship was strongly related in cases where only price and quality were analysed, the relationship was either weak or non-existent when introducing other variables, such as brand or additional information regarding the product or service in question (Jacoby, Olson & Haddock, 1971; Dodds & Monroe, 1985; Monroe & Chapman, 1987; Gardner, 1971).

An issue that seems to be prevalent from articles during this period is that there is a lack of definition regarding what quality is and its relationship with customer value. Quality has different definitions and meanings depending on the perspective and industry in question, where each perspective has its own benefits and consequences. (Reeves & Bednar, 1994). Within the context of marketing, there are two different types of quality that are often mentioned, perceived quality and objective or technical quality. The former is often tied to the attributes, functions or performance of a product or service and is often measured by experts (Grönroos, 1984). The latter is focusing on the interpretation of quality from the side of the consumer. An often-cited definition of perceived quality is that of Zeithaml (1988) as "… the consumer's judgement about the superiority or excellence of a product." (e.g Ophuis and Van Trijp, 1995, Desarbo et.al, 2001). The work of Zeithaml (1988) is often cited in customer value research in marketing contexts, mainly because of creating a comprehensive model of the relationship between quality, price and perceived value from the dominating perspective of means-end theory of the time. Leading to the conclusion that value perception of customers stems from a trade-off mentality between what the customer gains as utility compared to what they give up both in monetary and non-monetary resources, such as time or effort. Additionally, the value perception of the customers is influenced by a combination of intrinsic and extrinsic attributes of the product or service, which in turn influences a customer's perception of both quality and price (Zeithaml, 1988; Ulaga & Chacour, 2001). However, many researchers still came into issues with the subjectivity of customers with regards to

customer value. It was later tackled by the work of Holbrook (e.g Holbrook, 1999), in that the customer's perception of value is an interactive experience between the individual and an object, it is also relative to other available options from the customer perspective. Value can be intrinsic where the focus is the utility provided, similarly to Zeithaml (1988) but also extrinsic, where the value is in the experience of the customer upon consumption. Additionally, value can also be self-oriented or other-oriented, where the former is value only for oneself but for the latter it considers value provided for others in terms of micro scales as local communities or macro levels such as societies and countries (Holbrook, 1999). Thus, the work of Holbrook brought attention to new dimensions which focus on the subjective parts of customer value perception.

Although the body of literature is quite explorative with regards to this classical way of defining customer value, it is important to understand concrete ways in which customer value can be used in practice within development projects. One such way of putting the concept of customer value in practice is the Quality Function Deployment or QFD. The method is typically referred to as being used for guaranteeing that the needs of customers are the main drivers of the design and production of a product or process (Chan & Wu, 2005; Erdil & Aarani, 2018). This is achieved through the matrix which links the customer needs with requirements for product development, together with a comparison between an organisation’s offerings and its competitors (Cristiano, Liker & White, 2000). A visualisation of the matrix and its different parts is shown in Figure 1.

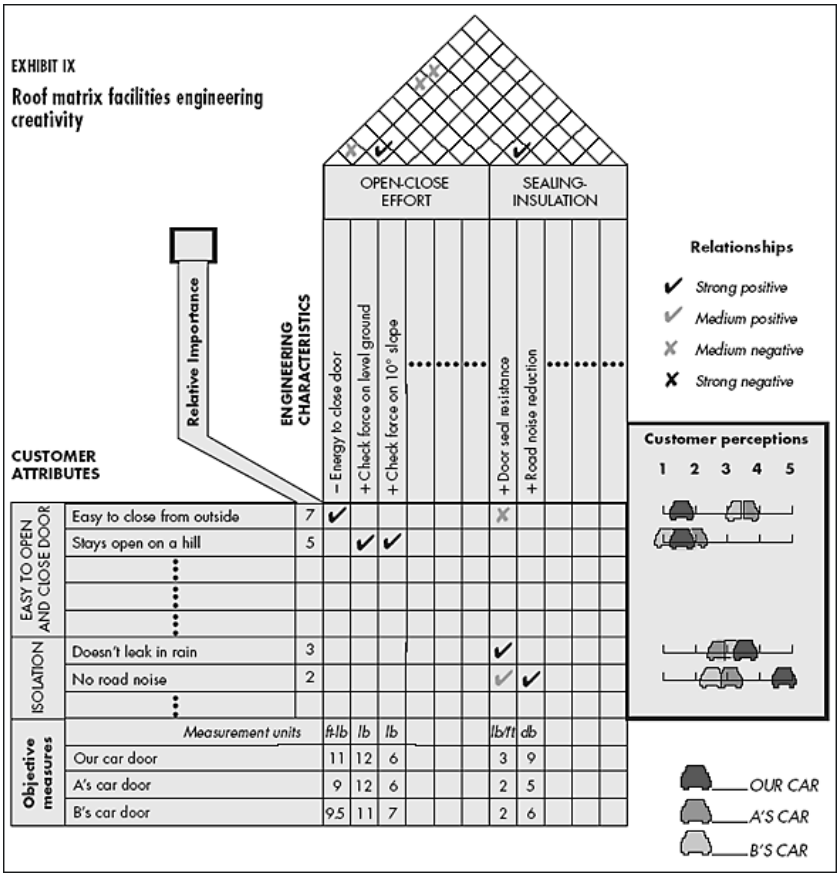


Figure 1. House of quality with its respective parts. Credit: Harvard Business Review

The visualisation clearly shows the "customer attributes" where the different identified needs from the customer perspective are presented. This section is also one of the parts which highlights the foundations of customer value understanding that it is based upon. For example, whilst the customer needs are often considered in the form of experiences, that experience itself is ultimately tied to the different aspects of the product, which is very similar to the customer value understanding of Holbrook (1999). Additionally, the organisation is setting the connections and their importance to the customer needs. Whilst it can be argued that organisations understand their offerings best, such approaches place an emphasis on the classical relationship between the organisation and the customer, where the former is a sole consumer, and the latter is a producer. It is important to highlight that the intention is not to demerit the method that has been useful and impactful for companies. But this may highlight a potentially interesting area to study, in that it is important to consider what definitions of the customer value concept are at the basis for tools and methods that are used during development. Another example

from the same branch of research as the QFD method, namely the quality management movement, is the Voice of the Customer (Bharadwaj, Nevin & Wallman, 2012). Primarily, the goal of the method is to increase the collaboration efforts between an organisation and its customers, specifically in product development endeavours (Melander, 2018). This is mainly accomplished by meeting the customer from an intended segment and gathering information through interviews, observations, and similar structured approaches, regarding the unique needs that customers possess (Matzler & Hunterhuber, 1998). Additionally, the needs are later categorised and logged for use as inputs in current or future projects, a keynote of this approach is that the expressions and formulations are unaltered as much as possible. The notion of gathering information to understand the customer needs is one of the most fundamental steps in the process of developing a product. Despite the adoption of the economic perspective of customer value, a new paradigm has emerged over the years which places a larger emphasis on broadening the collaboration between firms and customers, whilst introducing a new way of thinking of what customer value can be, referred to as Service Dominant Logic (SDL).

2.1.2 The Service Dominant Logic Perspective of Customer Value

SDL is a marketing theory which has mainly focused upon introducing a new perspective regarding the overall concept of customer value. It has first been introduced by the authors Vargo & Lusch in the year 2004, since then it has been adjusted and improved overtime as it has gained more traction because of its different paradigm compared to the Goods Dominant Logic (GDL) (Vargo & Lush, 2004). To begin with, GDL refers to the classical paradigm of value as mentioned in the former chapter, where the value is to be found in the product or object itself and in turn the value is being consumed by a consumer. Despite the highlight and existence of subjective parts with regards to how value is perceived within GDL, the basis is that value is a gains-sacrifice exchange with a clear distinction and roles as producer and consumer.

SDL focuses on a more intangible way of thinking, where a service is defined as the application of competences (knowledge and skills) for the benefit of another party (Vargo & Lusch, 2008). There are a few insights that can be extracted from this definition in

comparison to GDL. The definition places a stronger emphasis on the value creation process itself instead of the end result, it also takes a network perspective of the market instead of only the classic producer and consumer relationship. However, other aspects are not clear from the definition alone, for example, how is value seen within the SDL and what roles do customers and producers have within this logic. Value in SDL places the focus on the customer and their perceptions of value, also value only exists during the usage of a service by a customer and within the context it is used, this is known as "value in use" (Vargo & Lush, 2004). The value creation process within SDL, commonly referred to as value co-creation, takes place between customers, producers, and other potential partners, where an integration of resources, both tangible and intangible, is occurring to create mutually beneficial value (Eggert et.al, 2018). Thus, the role of the producer and customer shifts from one being a value creator and another being a value consumer, towards a facilitator and network partners, where customers also are interpreted as resources instead of just a target audience (Vargo & Lush, 2008). An important note within SDL is that although the definition and the paradigm itself might give the impression of pure intangibility, as the word "service" is often used, it does not necessarily mean that there are no tangible resources involved. A product or other tangible resources are viewed as bearers or carriers to enable the function or delivery of a service. A short example would be a car rental service, while the service itself allows you to rent a car to drive, the service itself would be useless without the involvement of a car.

One of the key elements regarding SDL is the concept of value co-creation and the unique view of integrating resources between multiple partners. However, while the theory is well established for the concept of co-creation, it is equally important to understand how. For example, one such part is the frameworks which condenses and gives clear action steps about how companies can implement SDL effectively at an organisation level. A notable work from Payne, Storbacka & Frow (2008) tackles the development of a framework, where their findings indicate that value co-creation is fundamentally about finding and exploiting encounters with customers and suppliers at different stages of the process. The authors emphasise the importance of developing new competencies at a micro level to handle relationships, which was also mentioned by Frishammar et al. (2019). Another example is the work of Karpen, Bove & Lukas (2012), which sought to link the SDL and

strategic concepts to thus create a conceptual model to guide managers towards a more service driven orientation. Their findings highlight six key capabilities that are shortly explained below:

- **Individuated Interaction:** Focusing on the concept of "value in use" and understanding both customer and network partner contexts and the ways in which resources are integrated, where the company fits in those processes and what outcomes and experiences customers desire.

- **Relational Interaction:** Take a more active role in managing relationships on emotional and social levels with network partners and customers to build tighter relationships.

- **Ethical Interaction:** All of this being done ethically such as to not take advantage of partners and customers.

- **Empowered Interaction:** Empower by providing and establishing the necessary support for customers and network partners to shape the content of offerings allowing them to interact on their own terms by enabling them to take a more active role in development.

- **Developmental Interaction:** Assist both network partners and customers in their knowledge and competence development.

- **Concerted Interaction:** The company's ability to facilitate the coordination and integration of services between customers and network partners.

These capabilities will be useful as a way to understand and explore the capabilities of the organisation in question of this thesis report. However, the authors mention that these are purely conceptual and empirical evidence of SDL in practice is highly sought after and a very high priority. There are attempts at exploring and finding out how SDL works in practice, for example, McColl-Kennedy et al., (2012) looked at the role and activities of

customers in a co-creation process of treatments in the healthcare industries. The authors concluded that the co-creation could prove more difficult in practice, because the patients in the studies interacted with different collaborators and integrated resources in unique ways even when presented with the same initial value proposition.

To summarise shortly, customer value is a deeply subjective concept where research is still being done to deepen the understanding around the concept. Two ways of describing the concept are those of the classical economic and service dominant logic perspectives. Key takeaways from the two perspectives were divided into four distinct categories. Firstly, Value definition which includes what defines value and how it is seen for each perspective. Secondly, Value creation includes what are the key ways in which value is created and who is responsible. Thirdly, Value relationship meaning the extent of collaboration between actors and customers when creating value. Lastly, Value delivery is about describing what the main means of delivery value are within each perspective. These categories and their application for the different perspectives of customer value can be seen in Table 1.

The classical perspective emphasises that customer value is seen as a trade-off between benefits and sacrifices, where the customer is a pure consumer of a product and an organisation is the producer, together with intrinsic and extrinsic attributes of products having a great effect on the perception of value. Additionally, it is important to acknowledge the fact that customer value is also experiential and relative for every individual. This perspective could be more prevalent as there is an established number of ways to apply the classical perspective. SDL instead places an emphasis on that customer value lies within the experience provided for the customer through a service which can potentially be enabled through a product. The value provided is also heavily context dependent upon the usage context of the service by the customer. Potentially meaning that the value for the customer can be hurt by applying a service in an unintended context. The creation of services that enable customer value are heavily reliant on the combination of skills and knowledge between customers and producers through co-creation, but also between suppliers and other network actors.

Table 1. *Summary of key categories from the two customer value perspectives*

	Value Definition	Value Creation	Value Relationship	Value Delivery
Classical Economic Perspective	Trade-off mentality between what customers gain for what they give up in terms of monetary and non-monetary resources. Highlights of customer experiences are acknowledged.	Inputs such as raw materials and customer needs are taken into a company to produce an offering which is then presented on the market available for customers.	A company is a producer which develops a certain offering which is later purchased and consumed. Customers are passive within the development process by only providing needs and testing products.	Value is often heavily tied to a product offering through its intrinsic and extrinsic attributes.
Service Dominant Logic Perspective	Value lies in the experiences provided for the customer through services during the usage of the said service, known as "value in use". Heavily dependent on within which context the service is being used.	Value Co-creation where companies heavily engage in integrating their resources and cooperate with customers and network partners to allow shape offering for a more active role in development.	Companies, customers, and other actors are part of a wide network to collaborate and focus on integrating resources to enable customers to shape their experiences.	Heavily reliant on services which are bundles of applied knowledge and skills to deliver intended experiences. Sometimes with the help of products but only to enable the service.

2.2 Innovation and New Product Development

2.2.1 Innovation Determinants

Innovation has been seen as one of the last and the most impactful ways to gain a competitive advantage in the uncertain markets of today to be ahead of other companies. It also creates new opportunities within the organisation which can be undertaken in the form of projects, customer value is an integral part of a project to fulfil the correct needs. This makes it important to understand what are the determinants that increase the likelihood of innovation to take place within a company.

There is an abundance of reviews regarding innovation management or different branches and its influences, one such review is that of Frishammar et al. (2019). Although it is a paper focused upon the auditing of innovation, it brings up the most relevant dimensions within innovation, those being “resources & capabilities”, “innovation processes”, “business model” and “culture”. Crossan & Apaydin (2010) go into further detail, as they break down “resources & capabilities” and “innovation processes” according to Figure 2. In accordance with the purpose of this article, the determinants of innovation at a managerial level will be investigated, for more details on a process level, see the related article of Ivarsson & Seing (2023).

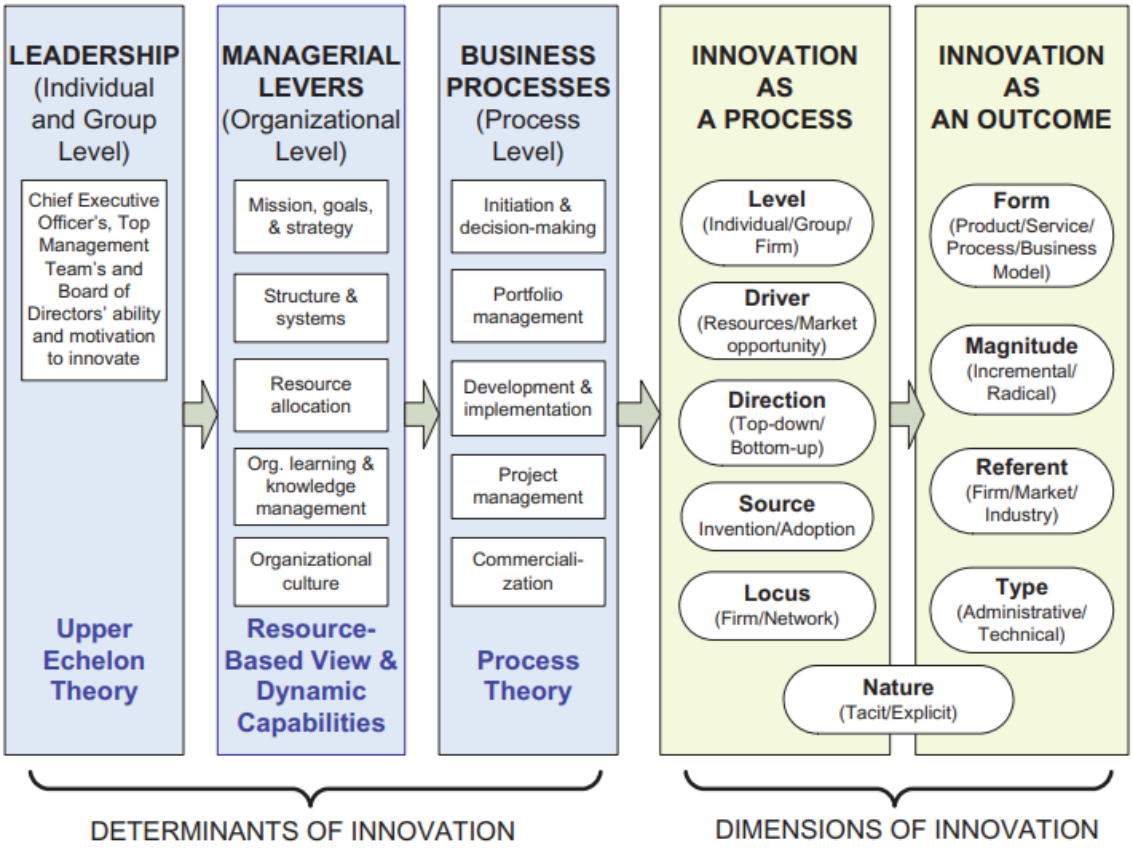


Figure 2. A Framework of Organizational Innovation. (Crossan & Apaydin, 2010)

The first determinant is that which refers to mission, goals & strategy. A mission typically refers to an often-written statement of an organisation's purpose, with the goal of communicating the intended purpose and values connected to that purpose across all areas of an organisation (Campbell, 1999; Inés et al., 2018). Although there are different ways to create a mission statement, a mission statement should seek to provide a sense of meaning or purpose for employees, while guiding the overall strategy and resource allocation endeavours of a company (Khalifa, 2012). Lastly, the concept of strategy is relevant in the context of innovation and is typically referred to as an innovation strategy. According to Pisano (2015), an innovation strategy is a bundle of actions, behaviours and policies that seek to align the multiple parts of an organisation to priorities towards similar objectives to support an organisation's business strategy. However, it is important to distinguish between the more commonly referred to as business strategy and an

innovation strategy. A business strategy will typically be concerned with the overall questions of "how to compete" and "where to compete" today, to establish a competitive advantage over competitors and is typically the responsibility of middle-level management. Additionally, a business strategy is also concerned about the future, by deciding what it wants to become and achieve, together with how it will get there (Grant, 2021). Organisations who seek to create an innovation strategy need to look at how its innovation efforts offer value to customers, but also how it will capture a fair share of that value and what types of innovations should it pursue to achieve all of this (Pisano, 2015). Lastly, it is important to note that an innovation strategy is not stagnant and should be worked upon and changed together with changes in marketing environments or organisation priorities.

Developing offerings often requires a group of people with different knowledge backgrounds to share and create new information and knowledge amongst themselves but also across organisations create a successful offering. Customer Value can also be shared as information as shown through the Voice of the Customer or Service Design methods. Thus, making it vital to understand how information and knowledge is shared, together with what drives the conversion between the two. Knowledge management can broadly be defined in two ways according to Gloet & Terziovski (2004). The first definition focuses on the benefits that organisations receive, in terms of new capabilities, superior performance together with encouraging innovation and enhancing customer value by formalising and increasing access to knowledge, expertise and experience. The second definition sees knowledge management as a bundle of separate but connected activities, concerned with the creation, sharing, measurement, storage and distribution of knowledge within an organisation. As this report focuses on an intra firm perspective, regarding how an organisation uses customer value in new product development. It is therefore important to understand how knowledge and information is communicated and how this impacts innovation within an organisation. Therefore, the focus will be on understanding creation and sharing of knowledge and information within organisations.

The most important distinction within the research stream of knowledge management is between information and knowledge. In the famous work of Nonaka (1994), he describes

information as a raw and unedited flow of messages, whereas knowledge is structured and controlled flows of information with the additions of beliefs and commitments of an individual. There are also mainly two types of knowledge, those being tacit and explicit knowledge as presented by Polanyi (1966), Tacit knowledge refers to experience, intuition or just in general tied to an individual's senses. An example of tacit knowledge is the skills of a carpenter, they cannot be easily transferable or explained in a short period of time. Explicit knowledge on the other hand is easy to be put into words or other forms of communication such as image or writing. A few examples are mathematical or chemical reaction equations, or just instructions for a recipe. It is important to remark that knowledge itself is not static and thus is not purely locked within the mentioned categories, knowledge is a dynamic process and shifts between the two categories through certain processes. A popular way of explaining the processes is that of Nonaka, Takeuchi & Umemoto (1996), where the authors argue that there are four main process that govern the conversion of knowledge:

- Socialisation (tacit to tacit): A process which is reliant on the sharing of experiences and creating new tacit knowledge as mindset and skills from already existing knowledge. The key for this process is experience through observation, imitation, or practice between individuals.

- Externalization (tacit to explicit): A process where tacit knowledge is more easily conveyed through metaphors and analogies and is often occurring during creative phases of a development project. This process is also vital for creating new concepts as metaphors help to relate multiple concepts and identify inconsistencies, thus leading to new thoughts. Whilst analogies help to find harmony between conflicting thoughts in a metaphor.

- Combination (explicit to explicit): It is a process which seeks to provide structure and organise a knowledge system, involving the sharing of different bodies of knowledge by exchanging knowledge through documents, meetings, telephones, or information technology.

- Internalization (Explicit to Tacit): This process can take place through experimentation, where explicit knowledge and other experiences are turned into tacit knowledge by putting those experiences to use. However, it is not the only way, another is through sharing experiences and successes to share the same mental models as other employees. Thus, internalising the knowledge within individuals, but also establishing the groundwork for that knowledge to develop into the groundwork for an organisation's culture.

Knowledge, a clear guiding vision and mission together with a learning oriented organisational culture incentivise innovation within organisations when done properly. But to succeed with innovation itself, it is important to understand how the development process within an organisation affects the likelihood of new ideas and innovation to occur.

2.2.2 New Product Development Processes

For a long time, one of the many ways for companies to gain a competitive advantage was to release new products before competitors , as this was generally linked towards more financial success (Cooper & Kleinschmidt, 1993). Additionally, it also leads towards the building of competences and skills within an organisation, which makes the company more resilient through periods of uncertainty (McCarthy et al., 2006). New products also lead to benefits towards society through technological progress and economic growth. Thus, there has been lots of interest within research of NPD processes, to potentially improve different aspects, such as speed or decision making, within a company.

The NPD process consists of a multitude of activities focused on specific stages within a firm to create new products or services from ideas to be released on a market (McCarthy et al., 2006). Typically, activities are categorised and take place within specific stages of the process, a few examples of important stages are idea generation, product development and testing. NPD processes vary between companies depending on factors such as company needs, industry or market together with what it is offering. However, there are general patterns in the way that NPD processes are structured. The most often used model for NPD within companies is a sequential model, an often-referred example is the Stage-

Gate Model. Key activities take place within stages in the model, in-between stages are gates, where “go” or “no-go” decisions are taken based on results and available information to either continue to the next stage or stop the project (Cooper, 2008). Sequential models are useful to quickly show the different steps that are to be taken and are often used as guidance. Thus, sequential models do not incentivise learning, creative activities, or the dynamics of innovation, but lead towards minimising risk and uncertainty in the innovation process through standardisation, because of this, sequential models tend to work well for incremental innovations but are lacking in developing radical innovations (Cunha et al., 2003; Bagno, Salerno & da Silva, 2017; McCarthy et al., 2006). Another type of NPD models is that of an integrative model, NPD within this view is seen as a complex activity which requires both information gathering, transformation and interpretation from many different areas such as market, technical, internal, or external information. Additionally, as the information required is more varied it also leads towards a stronger emphasis on having varied knowledge and capabilities within teams in such processes (Cunha et al., 2003). Integrative models focus more towards creating a well-rounded pool of knowledge, this emphasis also led to a larger focus on individuals within teams and understanding the effects of team dynamics in building effective and varied project teams. Thus, integrative NPD models typically can be seen as a knowledge creation process where current knowledge is used, and new knowledge is created which can be of use in the future. However, when the environment is changing rapidly, demands and information varies which also requires a certain level of adaptation. Flexible models seek to fulfil that adaptation by prolonging the duration of early phases, such as ideation and concept development, and focus upon iterative cycles and frequent testing, which leads to more feedback and knowledge internally (Cunha et al., 2003). A more familiar set of models to the flexible models is that of agile models. Agile models are described similarly as flexible models, as in making use of iterative cycles and reintegrating learnings towards a more defined final product, and such models also being used in fast changing market environments (Wynn & Clarkson, 2017). These types of models are therefore very dependent on taking action to gain more new insights to work towards the final goal.

To summarise, there are a variety of processes that can be used for structuring and providing guidance along the timeline of a development project. The type of process used for development projects can impact the determinants that influence the likelihood of innovation to occur. Stage-gate models, while providing great overview, guidance and control do not offer the support for dynamic interactions and learning activities to occur for innovation to foster. On the other hand, Agile and more integrated approaches can positively affect innovation determinants through quick implementation of the knowledge and insights developed from learnings which often resulted from testing often.

The established theoretical background explored in this chapter lays the important foundation with which to later analyse the data that has been gathered. The gathering and analysis of data together with implications of chosen methods is discussed in further detail in the upcoming chapter.

3. Methodology

3.1 Research Setting and Design

This study was partially initiated as a master thesis within the Integrated Product Design program with a specific focus on Innovation Management, during the period of 2023 at the Royal Institute of Technology based in Stockholm Sweden. The other part of the study was thanks to the collaboration with a major industrial manufacturing company, originally based in Sweden but now is active on a global level. Additionally, the company has had a long and strong history of developing products, from sewing machines, fridges, construction to defensive equipment and gardening and forestry equipment. It has shown time and time again that it possesses the capabilities and vision to adapt and innovate. Currently, the company has a general vision and mission to become more sustainable whilst still maintaining its historic identity as an innovator. One of the major ways in which the company has set forth in maintaining its innovation identity is through leveraging digitalisation and AI to create new offerings and enhance already existing ones. This thesis started as the company was interested in the value and potential of an AI-based offering. Whilst the case study for that specific offering was interesting, the author and company came to the consensus that it would be more valuable for the company to understand how its teams understand and apply the concept of customer value to make changes or incentives other mindsets to pursue their interests in product-service offerings. Lastly, this thesis was done in partnership with another team that consisted of Kajsa Ivarsson & Eric Seing (2023) as authors of the report; *The Road Towards Service-Integrated Offerings for Product-Centred Manufacturing Firms.* That report takes a broader focus upon processes and working towards achieving product-service offerings. The partnership involved collaboration regarding data gathering and multiple feedback sessions where suggestions for improvement were given regarding the contents of chapters in the report. Key parts of this thesis report, such as the theoretical background, data analysis and the results were done separate from the partner group.

3.2 Data Collection

The data used within the report is comprised of two major components. Firstly, it was the literature search which formed the basis of the theoretical background for the report. This was done by using keywords in the Google Scholar search engine. However, to improve upon the quality of the literature search, the Sixty-eight Edition 2021 of the Journal Quality List written by Professor Anne-Wil Harzing was used. A document which presents different scales that measure the quality of research journals within many research areas. For this report, the FNEGE scale was used, which goes from "1" being the lowest rating to "4". This scale was used as it is highly legitimate within the context of managerial research because of its support by 14 scientific associations within the management field. This scale and document were used to identify journals with impactful contributions and scientific reputation by choosing journals with a rating of "3" and above. Thus, journals from key areas to the purpose of this report, such as marketing and innovation management could be identified and were used together with the advanced search option in Google Scholar to find key articles from reliable and established journals.

The second major form of data gathering is part of the category referred to as qualitative research, using such form of research allows for a more open and involved process which can result in a more realistic depiction of the studied phenomenon compared to a more idealistic picture that more standardised approaches can create (Flick, Von Kardo & Steinke, 2004). This approach was placed into practice through the usage of semi-structured interviews, which was beneficial in the sense that it allowed for some specific topics with follow up questions. This allowed the informants to express thoughts that are relevant to the topic that would otherwise not be mentioned by the author through a more structured approach. The interview questions can be seen in Appendix B as Part 2 in the interview guide used in this thesis. However, one of the typically mentioned disadvantages of interviews is that they require large amounts of time and resources to both coordinate and travel to a location, but also for the analysis of the data later once the data has been gathered (Adams, 2015). The interviews were conducted in an online setting through the usage of online communication tools such as Zoom and Microsoft Teams, which eliminated the resource investments for both the author and informants

but also introduced unique implications for data gathering and analysis which will be mentioned in their respective chapters. Additionally, this approach was taken out of necessity as the participants were not available in a physical setting during the interview phase of this report. It does also have implications on the gathering and analysis of the data from the participants.

Firstly, there are unique implications with regards to preparing the digital interviews for a great experience. This was also because the participants were experienced in having interviews in digital settings and were thus familiar with preparatory work for the interviews, such as having stable internet connections, usage of both voice and camera functions whilst being in a relatively quiet environment to prevent distractions. All the steps are necessary according to Heiselberg & Stępińska (2022) as it would otherwise lead to large amounts of time being wasted which negatively affects the interview experience together with lacking data being acquired if not prepared correctly for both participants and interviewers. Secondly, the major implication regarding the usage of online communication tools is with regards to subtle communication cues, such as body language and eye contact, that can be limiting or lacking in a digital setting (Thunberg & Arnell, 2022). However, with the preparations and experiences of participants in digital settings and proper etiquette such as taking turns to talk the issues were minimised.

Many of the participants in the interviews were from three different parts of the organisation. Firstly, informants BD1, ID and RD were from a part of the organisation referred to internally as the AI-Lab, which is responsible with aiding and supporting internal departments with the labs specialised knowledge in machine learning and AI to improve already existing or new offerings, with a focus on early stages in a development process. Secondly, the informants PM1, PM2 and PM3 are part of the robotics department which is primarily responsible for the success of the organisation's automated lawnmowers. The three product managers are focused on the residential customers. Thirdly, this includes the informants, SD, PIE and ST, where the first two are professors at KTH and were interviewed for guidance into relevant topics for the theoretical background of the report, whilst the last informant was a service technician at one of the

organisation's stores based in Stockholm. Lastly, the overall information regarding the interview is shown in Table 2.

Table 2. *Interviewees from semi-structured interviews and formal meetings*

Informant	Role	Length	Type
Informant BD1	Business Developer	47 min	Semi-structured
Informant ID	Ideation Director	64 min	Semi-structured
Informant ST	Service Technician	40 min	Semi-structured
Informant RD	Research Director	60 min	Formal Meeting
Informant DPM	Digital Product Manager	79 min	Semi-structured
Informant PMD	Product Management Director	83 min	Semi-structured
Informant PM1	Product Manager	52 min	Semi-structured
Informant BD2	Business Developer	30 min	Formal Meeting
Information PM2	Product Manager	53 min	Semi-structured
Information PM3	Product Manager	51 min	Semi-structured
Informant SD	Professor in Service Design	66 min	Formal Meeting
Informant PIE	Professor in Product Innovation Engineering	53 min	Formal Meeting

The respondents were decided upon together with handlers from the company during regular scheduled weekly meetings, where general descriptions of goals and needed information was given. Handlers later used these descriptions to provide contacts where interviews were established. Although this approach led to some of the informants above, other informants were sought by asking for more recommendations at the end of interviews. However, there was a heavy workload period during this time at the company and thus no further interviews could be prepared. The implications of this were that more detailed information and representation was provided by informants from the robotics department as they possessed the correct knowledge compared to only two informants from the arboriculture culture team.

3.3 Data Analysis

Although the data has been gathered as mentioned in the previous chapter, it requires analysis for the discovery of useful insights and results. Transcription and coding of the interviews was performed in order to analyse the data, as it was overall qualitative data from the semi-structured interviews. However, both the transcription and analysis processes can be time consuming and complicated, but also daunting for inexperienced and unfamiliar researchers (Pope, Ziebland & Mays, 2000). The time consumption is a very familiar experience from previous projects with qualitative data for the author. Although a vital step, it was decided to minimise the time spent on transcription through the usage of automated transcription tools. The most convenient solution at the time of writing was the usage of the built in feature from Microsoft Word, which converted digital audio files into text documents. Although the accuracy can vary between programs, there are generally no greater problems than small grammatical errors that can be fixed easily afterwards. This was done through review passage of the transcriptions, where with the help of timestamps in the transcription and the audio recording, errors were solved. Lastly, the transcriptions were saved in a cloud storage for backup and access for authors of both reports.

The second part of the analysis included the process of coding the transcribed data. Coding simply refers to descriptive sentences or words to relate or assign a meaning to data in the transcription. Additionally, as remarked by Lester, Cho & Lochmiller (2020), the coding process should be seen as cyclical, in which the process starts off wide with lots of overall but not deep categories and codes. This is followed by the second phase, in which connections and relationships are sought after between the great amount of surface level codes from the previous phase. Lastly, deeper connections and relationships are sought in relation to theoretical concepts within research, thus connecting the experiences and expressions of informants with research. The author followed this exact process but with inspiration from Gioia et al., (2012), where the process is divided into three parts as “1st order concepts”, “2nd order themes” and “aggregate dimensions”. A visualisation of the results from this applied process can be seen in Appendix A.

3.4 Methodological Assessment

The main methodology of this thesis report was based upon a case study approach. Case study approaches are one of many research strategies that can be used to examine phenomena, particularly phenomena in singular settings and understanding the dynamics that affect the phenomena in question (Eisenhardt, 1989). Additionally, this approach is beneficial in situations where the phenomena is not yet well understood, or examining the “what”, “why” and “how” of a complete phenomenon (Voss et al., 2002). For this specific report, the phenomena were the practical usage and understanding of customer value inside an organisation. But also understanding what different factors affect this phenomenon, how it affects and why. This made the case study approach a good fit with the purpose and research questions of this report.

However, such an approach comes with important consequences that affect the results and their application in research. Firstly, one of the most common encounters is the amount of case studies. Single case studies, such as this report, tend to lack generalisability of results and conclusions, authors can also misjudge events and exaggerate available data but can provide deeper insights than multi case studies (Voss et al., 2002). Additionally, there is a misconception that single case studies do not provide as much theoretically as multi case studies, but single case studies do often rely on multiple case logic by analysing and comparing different cases within a similar context such as an organisation (Eisenhardt, 1991). The same thought process was applied for this report, by analysing and comparing different departments that were responsible for development, a more general application could hopefully be achieved. Additionally, the concept of customer value is a concept that is vital to understand as a company to deliver unique offerings for customers, especially with more variable markets and more unique customer demands together with a flood of new upcoming technologies. Thus, making this report important for other manufacturing companies to gauge their understanding and where to focus on creating or improving their capabilities to both understand and deliver customer value.

The partnership together with the authors of the other report brought interesting strengths to the data gathering process but also potential weakness to keep in mind for others that may be interested in adopting a similar approach. As previously established, the authors of the second thesis and this thesis performed their semi-structured interviews during the same sessions. The strengths of this approach became apparent during the analysis stage of the gathered data. Firstly, as a result of the joint interview sessions, it turned out that new insights have developed from the findings of each other's planned interview questions, perhaps because of the different perspectives, which were vital to the findings of both reports. Insights that may not have appeared if the joint interview session approach was not taken. Secondly, the interviews were performed in a semi-structured way to also allow for a degree of openness for the informants. This also aided the authors of both reports as the semi-structure approach allowed for unplanned questions to be asked which also led to new findings. However, these strengths are believed to require a basic understanding of each other's reports, purpose, and research questions to benefit successfully from such an approach.

A clear weakness of this joint data collection approach was that the interview sessions were more time limited to account for the fact that enough data had to be gathered for both thesis reports. In this case, one hour was divided into thirty-minute parts for each thesis. Thus, constraining the amount of time available that could have allowed for greater detail and examples to improve the depth of the acquired data. For this case, it was compensated with follow up meetings if not enough information was gathered in the initial semi-structured sessions. If opportunities for further meetings are not available, this places a stronger emphasis on planning, theoretical research and iterating upon the intended interview guide to be as prepared as possible for the limited time available for an interview.

Focusing on the quality of the employed research methodology, two often used criteria for ascertaining quality: reliability and validity. Yin (1994) has developed an adaptation of these often-qualitative criteria to the context of case studies. Validity was broken down into three types. Firstly, Construct Validity refers to the extent that correct measurement procedures have been done to analyse the concepts. This was done through heavy

collaborations and guidance from the project handlers, to make sure that proper qualitative approaches were implemented. But also the benefits and weaknesses of the chosen methods. Secondly, Internal Validity focuses on how certain conditions lead towards other conditions through clear causal relationships. Mainly done through careful coding of gathered data and constant discussion with the partner group to compare findings, thoughts and interpretations. Thirdly, External validity relates to the generalisability of the results beyond this report. For this case, the results are useful as they highlight important aspects, such as internal and external communication, that are beneficial for many organisations. Especially for organisations in manufacturing industries and other industries which are interested in taking their first transitional steps into product-service offerings. A summary of the activities that reinforced validity and reliability, together with where these activities took place can be seen in Table 3.

Table 3. Quality criteria adapted to case study context. Adapted from Yin (1994)

Test	Case study tactic	Phase of research
Construct Validity	● A clear and transparent chain of evidence is established in the report. Developed together with guidance from handlers from both KTH and Husqvarna regarding chosen methods and how they were used.	Research Design
	● Informants received an interview guide beforehand as preparation and insights.	Data Collection
Internal Validity	● Discussions and meetings for interpretations of coding results with a partner group.	Data Analysis
	● Used varied sources of information through experienced informats from different parts of the company.	Data Analysis
	● Using a structured method of coding as inspiration and guideline for coding.	Data Analysis
External Validity	● Finding links between established theory and results of coding from informants in different departments across the company.	Data Analysis
	● Research setting which is highly relevant for other manufacturing companies.	Research Design
Reliability	● Usage of structured interview guide that has been iterated upon.	Data Collection
	● Usage of multiple people as interviewers from partner groups to minimise observer bias.	Data Collection

4. Results

This chapter will first highlight what has been uncovered from the qualitative interviews with informants at the manufacturing company. All of this will then be analysed with the help of the established theoretical background and clarify how a company operates to apply the concept of customer value.

4.1 Findings

4.1.1 Perspectives of Customer Value between Departments

One of the first findings that became apparent from the interviews was that there were similar definitions regarding customer value between teams from different departments in the company. Additionally, Informant DPM built upon this view by mentioning that it is not only about what the value for the customer is, but also looking at other dimensions of value and how they impact the value delivery to customers. An explanation of this was that although there is a clear overall mission and vision from higher ups, teams interpret these in their own ways depending on what the teams are capable of. Thus, leading towards more teams working to preserve current offerings to bring in profits, whilst other teams, like the one that Informant PMD is responsible for, work with projects and offerings to move the company towards the future. To clarify, this sentiment was expressed to not be a problem but that it was expected, as the company needs to innovate at the same time as it remains profitable.

"It's not that it's 2 kilos, it's that it is super ergonomic to use, and those things we are willing to pay money for and add more because that is kind of the pillar of what we are doing here and the reason for choosing Husqvarna." - Informant PMD

Informant DPM expanded upon the definition mentioned above by highlighting that it is not only the careful and quality design of attributes, but also other aspects that affect the customer value.

"We try to find products or try to create products that can really create value for the customer and hopefully it's in a way that they haven't experienced before." - Informant DPM

An example as one of the ways that a new experience would be introduced in the future is that of an interface that is readily available on the product itself that showcases information about the product in an easy and convenient way, even though it might be the same type of product as previously offered. This definition of customer value is also used when working with digital components of a value offering according to Informant DPM who is responsible for digital solutions and connectivity. However, it was unclear Informant PMD mentioned that although it is generally easier to implement features in digital offerings, such as apps, it is difficult to identify what the true value is for the customer. The risk being that customers will not engage or will engage with it but choose to not do so because of previous bad experiences. A separate informant, namely informant DPM, who is the head of so-called AI Lab within the company, mentioned that from his experience, there are complexities that arise from working with services. One such complexity is the idea that services are composed of smaller services that individually provide some value or none, but the overall value may be much greater when the smaller pieces are put together to operationalize one service in particular that solves more problems. Thus, the complexity being the uncovering and pairing together the right services to solve the right problems. Informant DPM suggested that one of the reasons for this complexity is from the company's roots in product development, but also from a lack of knowledge, that if not readily available inside the company is explored outside the company through different collaborations.

"We might reach out to students or other means to kind of build on that knowledge, because that could be a big barrier for change that we don't know." - Informant ID

Three of the remaining informants, namely Informant PM1, DPM and PM2, are part of the product manager team within the robotics division of the company, thus working with both physical and digital parts of the automower offerings. The focus of Informant PM2 responsibilities are increasing customer value through digital services while Informant DPM is focused on the product side of the offerings and its features. According to both informants, the understanding of customer value is based upon an internally developed strategy. This strategy is heavily anchored around user goals that have been discovered

through learnings, findings and research done by the department since its release of their successful core offering. The four user goals are expressed at an abstract level and from the user's perspective, with a particular focus on the experience the customer should encounter, as follows; "I have more time", "I feel in Control", "I am proud of my lawn" , "I feel smart". User goals are to be achieved in all offerings developed by the department through its core offering and other offering aspects being driven from trends, internal roadmaps, and the missions & goals of the company. These user goals are then broken down to be more specific for each development project, thus providing an overall guidance for the teams in the department for what to fulfil and strive for. The Managers also have the expectations that by pursuing the mentioned user goals it will strengthen the company position as an innovator, improved customer relationships and a leader within the product category of the offering. However, this focus upon the customer goals has not always been that way, the department initially was focusing on the product and building around it by taking advantage of new opportunities through digitalisation.

"It had been a focus around the product, being the digital experience was an extended arm of the product. So, all the features that were in the mower were also represented in the app, where you can control it but then it also became a limitation for developing further because it was so connected to what we do in the mower we are supporting in the App. But when we started to shift our focus from having the product in center and to the user in center and the lawn in center. We could see a lot of other opportunities that we could do." Informant PM2

Although the strategy regarding customer value for the department has shown to be clear and easy to understand from the informant's perspective, a constant challenge is the communication of the strategy throughout the organisation. Namely, making sure that the priorities of other departments are aligned with the priorities of the robotics department. But also, the usage of how the robotics department uses values within the development projects and places the customer and their values in the center compared to the product. A way to improve the communication is the usage of metaphors to make communication more memorable and relatable.

"Key is to find those metaphors or find how to best communicate this message and not only doing the PowerPoint presentations and the talk about the user values over and over again, but finding good examples that stick." - Informant PM3

Informant PM2 also expressed the similar ways of improving communication and clarified with an everyday example or experience that a majority can relate to.

"We have a metaphor when it comes to the app that we should become a PT for lawn care, a personal trainer. We used to be like a gym card where you can have everything you need to succeed because there's a lot of settings and things you can do with the app. But we want to take you by the hand and help you. So, we want to be a PT for long term care, to give you personalised and customised help to succeed with your tomorrow, but also with your lawn." - Informant PM2

An often-encountered challenge related to customer value from the experience of Informant PMD is that individuals relate the value of an offering directly to its features. A solution undertaken by the same informant was the recruitment of individuals with more business or entrepreneurial backgrounds, as this challenge was typically encountered in teams with members that have predominantly engineering backgrounds. Similar thoughts have been expressed by Informant DPM, from his experiences with marketing and sales departments where an overemphasis is placed upon the features when communicating or selling an offering to the customer instead of the value itself.

"I mean, it's so easy for an engineer to think about values as fixed. I mean it's the change in speed, it's the weight." - Informant PMD

A similar sentiment was shared by Informant PM3, mentioning that there can be an overemphasis on hardware and features that can limit new potential.

"From a user perspective it shouldn't matter at all if the value is delivered through the digital service or through embedded software or through mechanical part, we're trying to communicate around the value, but there I think it might also be a challenge to get past talking about a certain hardware or certain feature and rather talk about the benefit and the value."
- Informant PM3

4.1.2 Processes and Ways of Working

To start off with the processes of the teams, both informants PM2 and PMD have initially mentioned how their distinct processes aid in the work with customer value. In the case of the robotics department, their process is highly iterative and visualised as the infinity sign, that it is comprised of two main stages: discovery and delivery. During the discovery phase, the goal is to confirm that the project in question delivers upon the user goals to the consumer. This is followed by the delivery stage in which the offering is pre-released or tested with customers. Additionally, informant PM1 mentioned that customers are typically involved with projects at early stages to uncover and understand customer needs, together with later stages of development for testing and feedback where internal testing is also performed. The whole process mainly takes place within three-month increments, where prioritisation has been done depending on the developed roadmaps and degree of fulfilment of the intended user goals in the department's strategy. Daily scrum meetings and sprints are normal ways of working within the previously mentioned phases. A rather new part of the process which the robotics department is working on is the feedback loop, which is currently being done through the close tracking of reviews, surveys and ratings which are the measurements used for gauging the customer experience. However, the department is still learning how to improve and work more effectively with the inclusion of the feedback loop. The arboriculture team mentioned similar working practices through agile and iterative practices, which enables the team to test and validate different aspects of a product or offering before investing large amounts of resources. Another benefit that stems from the frequent testing and iterating nature of the team's process, is that it brings common understanding regarding what delivers value for the customer within the team. For example, beyond just validating and confirming that aspects of a product are functioning properly and delivering upon the promised value.

"What we are trying in my team is to be more like a startup. It must be we're going more to have a look … to do small sprints to test instead of spending 10 million SEK on a two-year product project and I think we must be more agile." - Informant PMD

To make use of the processes highlighted above, the inputs of ideas and how they are chosen is vital. Starting with the robotics department, informant PM1 mentioned that part of the work as product manager includes identifying potentially new ideas. The inspiration for ideas has been expressed as being mainly internally driven and about matching internal competencies and resources with the roadmaps developed from the vision and mission of the company.

> "I mean, we have a kind of a vision of where we want to be and then depending on what kind of technological capabilities we have on doing this and on which products do we want to have this first."
> - Informant PM1

Similar thoughts have been expressed by Informant PMD of the arboriculture team, highlighting the relationship and degree of collaboration between the company and the customers.

> "We don't get any suggestions from our customers. Of course, there are exceptions where the customer actually thinks beyond the box more or less, but it's up to us, I think to very much see the needs and interpret those into ideas that we then give to primary and their gross list to make maybe look into more and make concepts and so on." - Informant PMD

Once ideas have been identified, the robotics department compares the potential of customer value of the intended offering with current offerings in accordance with their strategy. Once the inputs in the form of ideas have gone through the processes described earlier, a key deliverable is produced, mentioned as market-user requirements. The requirements start off initially as needs expressed by customers at a general level, which are gathered mainly through marketing and sales departments, but also other channels such as the global and large network of dealers that the organisation has access to where customers provide feedback. But also, where dealers engage in knowledge sharing and supporting customers all around the globe. Information gathering techniques often include surveys, questionnaires, and face-to-face interactions with customers. The market user requirements which are expressed with as little technical information as possible to focus more on the benefits for the end users. Lastly, it was also mentioned that information about the customer segments, their unique contexts and experiences is also

vital in creating these market user requirements to represent the value for the customer as accurately as possible but also to ensure that the information is accurate.

"Even an idea that doesn't lead to a successful launch maybe is a seed for something else or like leads to another product getting born that wasn't in anybody's thoughts from the beginning, so everything has its place." - Informant DPM

"That is what we're doing with the market use requirements, we tend to try to leave the technical data that is not the customer value, it's the benefit for the user." - Informant PMD

"So, our biggest or most important kind of delivery in a project is something that we call market user requirements. So, it's basically a document where we sign off what this is. This is what we want and then there's like a technical specification from R&D... that's our biggest and that's early on in the project that we do that. Before we actually get the project approved." - Informant PM1

The last step is to then take these ideas that have been converted into projects to decision committees, where the projects are represented by the value for customers and market-user requirements to describe the offering and its contents. These committees are composed of multiple experts from different departments of the company, this is a general activity which takes place outside the teams' boundaries. If an idea is not considered to be good enough to invest for a fully-fledged project, because of these activities, the ideas are still considered valuable and could prove useful in enabling future ideas or projects because of the developed learnings along the way.

"We call them project briefs: Essentially one pager, where we sort of describe: what's the problem, how it's solved today, what's the value, if we can solve it? If we would do a project; How we will go about it? How long do we think it'll be? Our possibilities for doing patents... Things like that, and we try to sort of summarise that and together with the stakeholder." - Informant BD1

However, if a project has been approved by the decision committee it is then taken further as an input into the New Product Development process of the company, internally also referred as Product Creation Process. This process is heavily based upon the Stage-Gate Model with gates representing specific key decision points in the process, such as specification of product or investment of resources. Followed by the phases themselves

which take place in between the gates and give an overview of the core activity taking place. Informant PM1 mentioned that this approach is highly effective in the case of the organisation, as projects involve large amounts of internal stakeholders and makes it easier to divide and coordinate the work between departments. Additionally, approved projects are large investments for the company, thus a toll gate approach allows for more control and structure to minimise unnecessary resource spending, according to informant BD2. Collaborations have been mentioned by informants from both teams, which is not a traditional way of working. Previously, this was beneficial for the company because the different departments and teams could focus on delivering on the parts of a product which suited their knowledge and capabilities the best. This led to successful results when delivering products, but according to informant RD, the company realised that a variety of knowledge is more beneficial for developing services, and thus led to more collaboration between departments on service-related projects. Not all these collaborations are internal, informant PMD mentioned how they interact with different companies in early phases of projects to develop concepts, collaborations also involve suppliers to ensure smooth operations and understand the capabilities of both sides. whilst informant BD2 spoke of collaborations with other AI divisions and sharing knowledge on AI and machine learning subjects. Interestingly, when the informants from both teams were asked about collaborations together with customers, the responses were generally that customers are involved but only early in a project's timeline and testing phases late in development.

5 Analysis & Discussion

To understand the different interpretation of customer value and the practical implications this has on the development of novel offerings; the gathered data will be analysed for each respective research question with the help of the established theoretical background.

RQ 1: What are the perspectives regarding the concept of customer value within different groups during a product development project?

The definitions of the customer value concepts within two different parts of the same organisation appear to be quite similar. Starting off with the arboriculture team that informants PMD and DPM are part of, the definition for this team emphasises the experience of the customer during its usage, which aligns itself with both the notion of extrinsic value mentioned by Holbrook (1999) but also the SDL perspective by Vargo & Lusch (2004) with the concept of "value in use", thus potentially connecting the GDL and SDL perspectives around that customers experience is a vital part of the overall customer value. The interpretation of the robotics department was expressed in the form of a clear four step strategy, where a similarity emerges between the mentioned "user goals" and the emphasis on experiences from the arboriculture team. In the case of the strategy for the robotics department, it highlights two important things. Firstly, new mental models have emerged through the knowledge developed by insights and findings was a vital key in the development of the department's strategy, highlighting that knowledge sharing is important to innovate as mentioned by Nonaka, Takeuchi & Umemoto (1996). In this case it led to a strategic innovation which in turn led to the interpretation of customer value in a new way, by not always developing services as a support of a product, which opened new opportunities for the robotics department. However, there have been implications of this innovation, as a difference in mental models has complicated the communications both internally in the team but also across different departments in the organisation. This communication was handled through the process of externalisation as explained by Nonaka, Takeuchi & Umemoto (1996) using metaphors to communicate the developed

tacit knowledge from the robotics department towards more explicit knowledge for other employees. Overall, there is a similar understanding or interpretation with regards to customer value between the teams because of knowledge and learnings developed along the way. However, an equally important part was the processes and how the concept of customer value is used within those processes.

A more product centric approach, where the careful and quality design of intrinsic product attributes enable quality experiences. But also, a more service centric approach, where an intentional usage of a personally customised service within a correct context delivers the unique experience and therefore value to the customer. The main driver for these approaches seems to be the unique capabilities that each team possesses. This has led to complications regarding internal communication which can inhibit the spreading and adoption of new knowledge. In this case, knowledge about

RQ 2: How is the concept of customer value used within the product development process of the organisation?

To better understand how customer value is applied, it is important to first give a short overview of the processes from both departments. Initially, both teams have mentioned the adoption of iterative processes as their main ways of working with development projects. The arboriculture team applies Agile principles in their processes to iterate and test often to prevent large amounts of investments like the mentions of Cunha et al., (2003). Another effect of this iterative approach was that it ensured that all team members had the same understanding when it came to what was valuable for the end customer, similarly to the effects of an innovation strategy as mentioned by Pisano (2015) but at a team level instead of an organisational level. In the case of the team from the robotics department, the described process also emphasised the iterative nature and learning focus of their “infinity sign” process which was divided into discover and delivery phases every three months. An iterative approach such as their “infinity sign” led to a culture that promotes learning and knowledge building that led to the department’s strategy around user goals. Thus, emphasising the importance of innovation

determinants such as organisational culture and knowledge sharing as mentioned by Crossan & Apaydin (2010).

The processes described earlier appear to be used at early stages of development, where the early development work is focusing upon creating the market user requirements mentioned by informants from both departments. This will be followed by a decision board to fully commit towards a project. Initially, customer value takes the form of pure information gathered from customers, partially by the marketing and sales departments, employing similar approaches as the Voice of the Customer described by Matzler & Hunterhuber (1998) but also the goals of the approach. Those being user centred from the company and having the customer needs drive the development, but also in practice through interviews, surveys, and observations. The other part is done by the departments themselves, i.e the feedback loop of the robotics department or by experiencing customer contexts as the arboriculture team. This information is then used as an input in the processes described earlier to create the often mentioned "market user requirements" by the informants, which in turn leads to a project suggestion or brief. However, from the SDL perspective, this approach continues to incentivise the classical relationship between the consumer and producer Vargo & Lusch, (2008). Thus, potentially excluding unique knowledge that customers can possess for new ideas or improvements in conjunction with multiple network partners that could lead towards new innovations and competitive advantages. However, working together with customers and other potential network partners requires new capabilities, knowledge, and resources (Frishammar et al., 2019; Payne, Storbacka & Frow, 2008). One way of understanding the required capabilities is through the service driven orientation capabilities suggested by Karpen, Bove & Lukas (2012).

Informants from the two separate departments seem to be aligned, at least theoretically, in their understanding of the concept of "value in use" where the value of the customer is heavily connected to the context in which the offering is used with the goal of delivering the intended experience. Practically, the robotics department develops products, often to enable a service, together with service offerings, whilst the arboriculture team seems to be heavily reliant on products but are working with incorporating digital technologies in

their products. The teams seem to have a good understanding of their customers and what they want and understand that their offerings help enable the smooth experiences that customers want. This may indicate that the "individuated interaction" capability may be present on the surface, but the capabilities may not be in place yet for heavily personalised experiences overall, where a potential roadblock will be the complicated ways in which customers engage with other network actors as mentioned by McColl-Kennedy et al., (2012). Speaking of customers, the "relational interaction" seems to be established by the company through its dealerships where contact with the customer where their questions, feedback and support is given. However, when it comes to the network partners and the organisation, it seems to be purely business and knowledge building oriented, which can lead to lacking emotional or social interactions. Dealerships and suppliers seem to already be some of the important network partners that the organisation already engages with regularly to ensure that their operations on a global scale run smoothly. Thus, the company already possesses the capabilities to achieve "developmental interaction", but a potential for improvement could be in supporting customers to a higher degree. The "empowered interaction" capability is also quite limited from the informants' responses, mainly because customer interaction is mainly in the traditional sense as understanding needs to then be used as inputs in development projects and test the results towards the end of development. A similar approach seems to be taken with regards to collaborating with other companies, the collaborations mainly happen early in the stages, typically until a concept has been formed. Thus, the organisation in question could improve upon this capability to allow both customers and network actors to have the opportunities to interact even during the development of an offering, not just at the early beginning and close to the end. However, from the data gathered no clear analysis can be made about the "concerted interaction" capability, other than that this capability could be underdeveloped as there is no indication that the organisation engages with other network partners beyond suppliers, especially trying to coordinate any sort of communication between customers and other network partners.

6. Conclusions

The purpose of this report was to investigate the different interpretations and applications of the customer value concept by two separate departments in a manufacturing company. This chapter will aim to summarise the conclusions that have emerged from the analysis done on the gathered data for each respective research question of this thesis report. Followed by highlighting potential theoretical contributions and managerial implications from the presented conclusions.

RQ 1: What are the interpretations regarding the concept of customer value within different teams throughout a development project?

The main parts of this study were the different teams from separate parts of the organisation. It became evident that both the robotics teams and the arboriculture team were very similar in terms of their explanations and understanding of the customer value concept. The similarity being that both parties expressed that value for the customer lies in the experiences delivered to the customer which are enabled through the available offerings. However, communication issues happened for teams that did not have similar ways of interpreting customer value as the teams that were part of this thesis report. Thus, companies should seek to understand the different ways that teams comprehend and interpret the concept of customer value. This is important as different conceptualisations of the customer value concept could give rise to different mental models which can introduce internal communication issues. Introducing new communication techniques, f.e metaphors, to simplify the communication of subjective topics such as customer value can be impactful in companies where cross-functional collaboration is vital. Differences were found with regards to what the respective teams believe to be the main way of delivering value to customers, on one hand the arboriculture team took a more product centric approach, where the careful design of product attributes enabled quality experiences. Whilst the robotic teams took a more service centred approach, where an intentional usage of a personally customised service within a correct context by the customer, delivers the unique experience and therefore value. A potential driver for these

beliefs of delivering value could be the capabilities available within the teams, thus the teams are working towards offering value in accordance with their respective capabilities.

RQ 2: How is the concept of customer value used within the new product development process?

The concept of customer value was typically used during the same stages that involved customer involvement, first during early stages of pre-development, while the offering is still an idea state that is yet to be considered for development. Distinct processes were used by the teams during pre-development, but the overall goals and visions for the processes were the same, to test and fail fast early on to explore as much as possible and gather learnings. Both to better understand the customers and their unique contexts but also to minimise the likelihood of making costly investments into projects that may not pay off in the long term. Thus, iterative, or cyclical processes were used to structure and achieve the previously mentioned activities and goals. It is important to remark that these processes were supported by an organisational culture which sees failures as opportunities of learning to thus create new knowledge that could prove valuable for the development of future offerings. These same supporting elements are recommended to be in place at other companies to successfully enable such processes to be successful.

A second instance of the customer value concept in practice was related to the testing phases, where different aspects of an offering are tested both in the company and with customers. However, service dominant logic suggests that empowering and enabling customers to collaborate with companies and network partners is a key element of developing successful service offerings. Additionally, it is unclear why the company does not participate in deeper collaborations with customers, beyond testing and understanding their needs at early stages. This can lead to gaining further insights and ideas from customers as a new source of knowledge and learnings, but this will require key capabilities to leverage. In the case of Husqvarna, the company already possesses a foundation for relational capabilities which it can expand upon, but it may be difficult for other companies to develop the necessary capabilities.

6.3 Managerial Implications

The findings of this thesis report have highlighted important implications to consider for managers at manufacturing companies. Firstly, it is important for managers to identify the degree of collaborative engagements, both internally and externally. The findings suggested that a manufacturing company engages in heavy internal collaborations, especially for development of product-service offerings. These internal collaborations often involve teams across the different departments of an organisation, which can have different ways of defining and implementing concepts. One such key concept in this case was customer value when developing product-service offerings. Product-service offerings include service elements which can be new and not well understood by a traditional manufacturing company, especially in terms of what value service elements offer and how it is delivered together with other product elements. Thus, managers need to identify how different departments in the company interpret customer value. But also, to establish activities that inform departments of their ways of thinking about concepts such as customer value at a basic level. It is important to highlight that these activities should not seek to force change upon a department's mental models, but to instead develop internal understanding of mental models between departments. Secondly, managers should also seek to leverage collaborations with other organisations, academia, and institutions both within and outside their present industries. This brings new perspective, knowledge and resources which can be extremely beneficial for manufacturing companies, especially small and medium sized companies who may be constrained. Additionally, such collaborations can be even more beneficial if expanded during the lifespan of a project not just in early development stages, as more knowledge is shared which introduces more opportunities and relationships are strengthened while the capabilities developed will be useful long term.

However, even if similar mental models are already understood and collaborations are leveraged, differences arise in the ways teams and departments work during development. Thus, it is equally important for managers to develop internal understanding of processes and capabilities between departments, together with the impacts of collaborations and mental models on the overall development process. Lastly,

managers should also investigate re-designing current development processes towards more iterative approaches. Both to allow for certain amounts of flexibility, to both adapt to new information and more easily include the aspects mentioned in the previous paragraph, and promote learning.

7. Limitations and Future Work

Limitations were placed upon the report for it to be more focused but also because of time constraints. The first limitation was the inclusion of only the two departments from Husqvarna that were part of this study, although that more teams from different departments were potentially available but would have expanded the requirement on both theory and data analysis which did not fit with the time constraints of this study. Additionally, the respondents only included managers within the different departments, considerations should be made in future studies to include other employees and thus gain a varied number of perspectives. Another limitation was that the concept of customer value is heavily reliant on the customer aspect and their interpretations, which was not considered for this study. This should be considered as a vital aspect to include in future studies, specifically the customer's interpretation and value estimations concerning product-service offerings.

Future work to build upon or expand upon the practical implementation of customer value and its interpretations should also include a couple of key aspects beyond those already mentioned above. One such aspect is that of digitalisation as it is a fast-growing trend with great impacts on all types of firms, especially for manufacturing firms because of their traditional product-centric approaches. More specifically, how digitalisation in product-service offerings affects customer value for customers and how can this value be delivered, and a fair share be appropriated by the company. This is related to business models, where investigations of relationships between customers and product-service offerings can affect an organisation's business models could be beneficial for both researchers and practitioners. A good start for analysing and exploring the concepts mentioned earlier is to perform similar studies, but through a multi-case study either at companies within the same industry but also other industries, to expand the generalisability of the findings. Future studies should also pair qualitative and quantitative approaches such as interviews, observations with surveys to gain richer data that can provide deeper insights.

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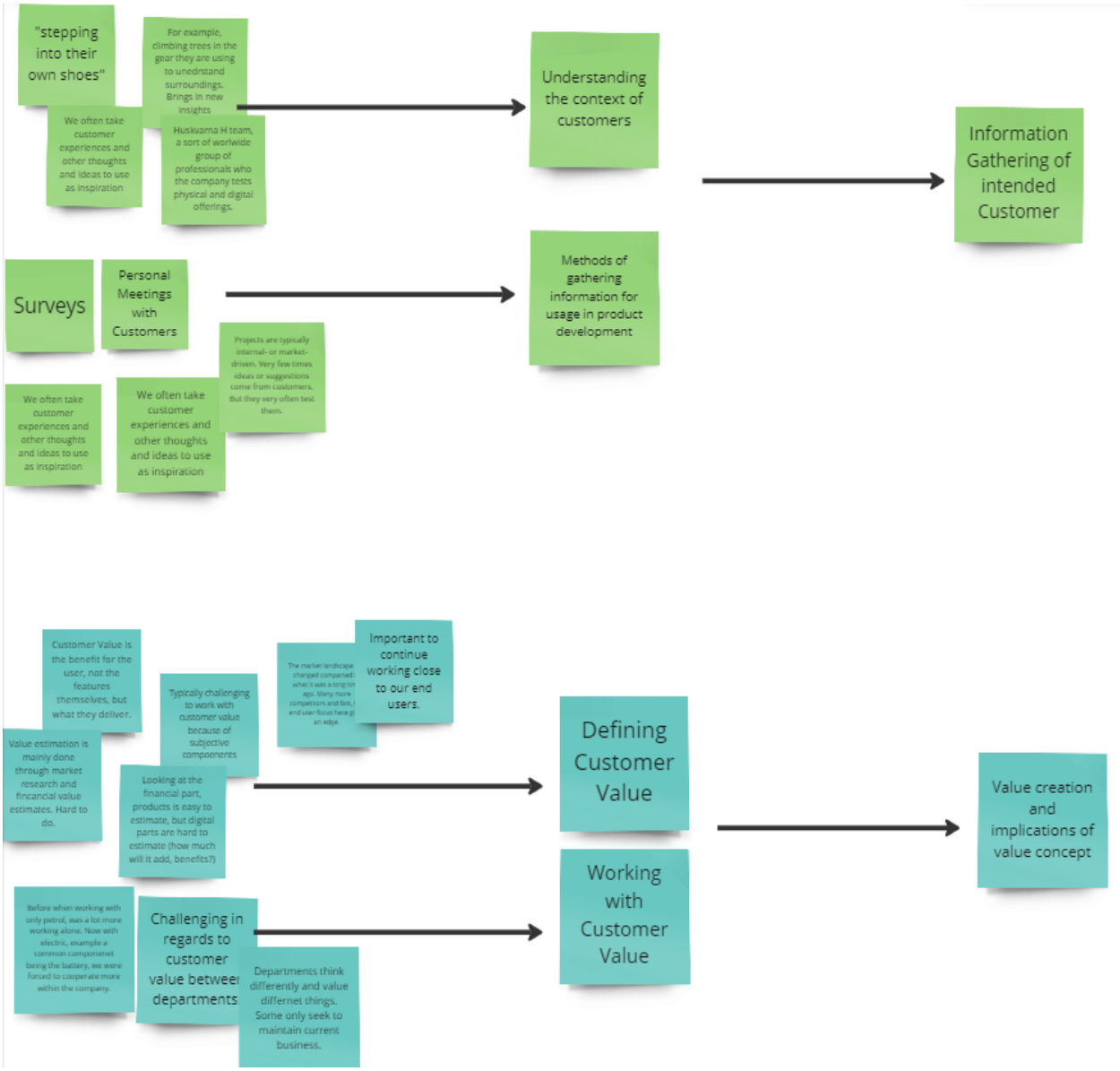
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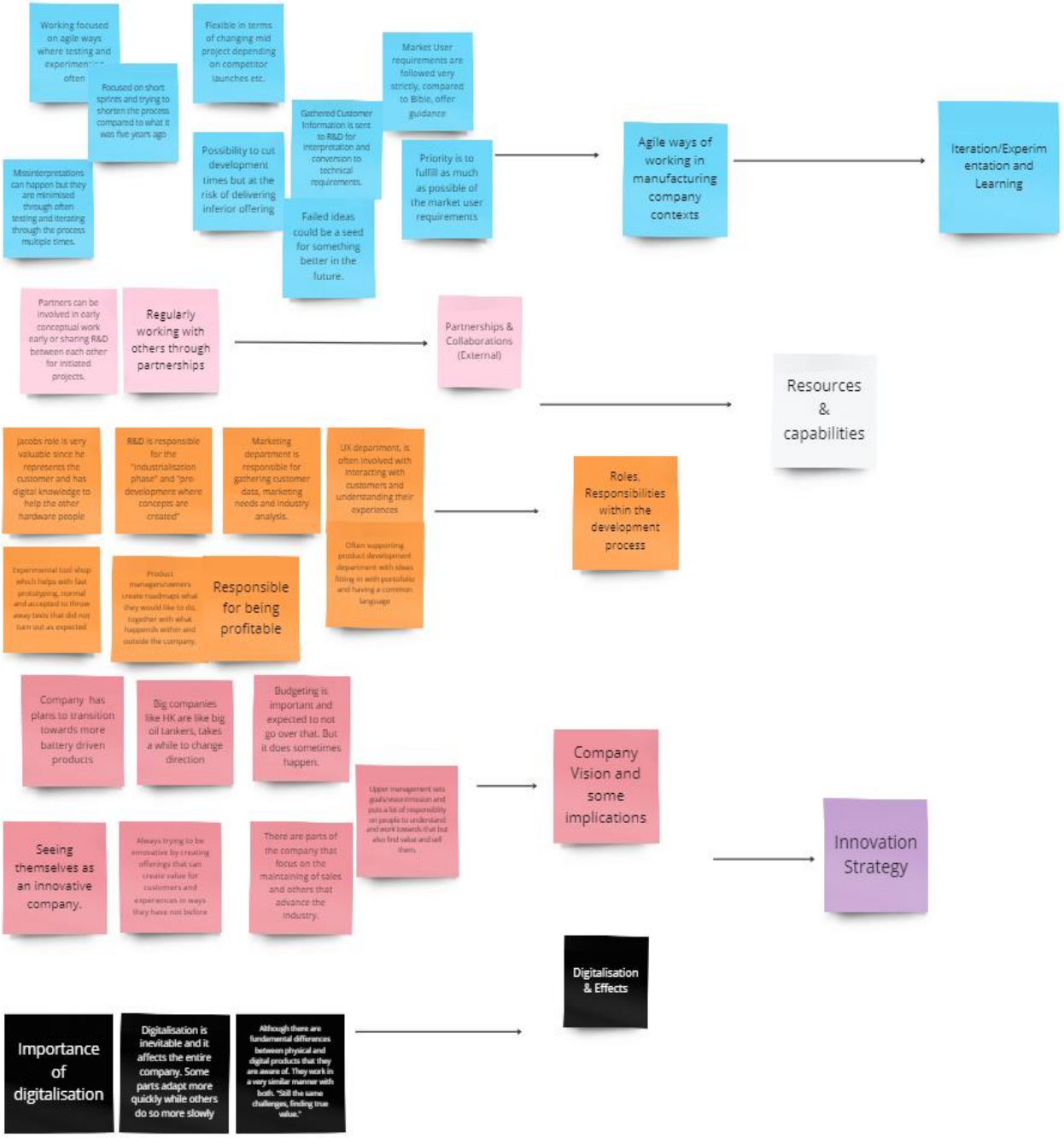
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Appendix A - Coding Example





Appendix A: 2(2)

Appendix B - Interview Guide

Informant(s):
Date:
Time:
Department:
Format:

❖ **Introduction**

- Tell us about yourself, who is [insert name] and what is your role at Husqvarna?

PART 1 - Innovation Process

- Can you briefly explain how the current product development process works?
    - What about the development process for product-service systems?
    - Any examples?
- How do you work with the customers? Are they involved?
    - Which part of the product development process, the early, mid or end?
    - How do you receive feedback?
        - And how do you use it?
- When developing a product, can you briefly explain how and what parts of departments work together within Husqvarna? Outside Husqvarna?
    - Examples?

- Currently, you have some interesting projects such as ***Husqvarna Fleet Services*** and ***Tools for you***. Have you been a part of these developments or is knowledgeable enough to give us any information about how their Product development processes look/looked like?

    - If not, could we get in contact with them?

- Do you have any other experiences closely related to this subject, concerning any product and service offerings? (it does not have to be a finished product)

- Do you currently have any ideas of clear issues that can be run into when implementing product service systems in Husqvarna?
    - Example: Are there any departments that could be sensitive to the change related to the transformation?

- What does the decision-making process look like? Do you use a method?
    - Does it change depending on what "phase" or "stage" the product is currently in?
    - Is it any different to a product-service offering process? Do you have any examples from projects you were part of?
        - If not, could we get in contact with them?

- Do you weigh a product on criteria? [Specific Question]
    - Which criteria?

- How do you decide if you need to kill ideas?
    - Early on as well as later on?

- Do you have any other experiences closely related to this subject, concerning any product and service offerings? (it does not have to be a finished product)

- How do you determine when a product is ready for launch? Do you have any examples?

- How does that change to services? (Fleet Services or Tools for you) Or other projects you have been a part of?

PART 2 - Customer Value

- How do ideas come in to be considered for development?
    - How are these ideas presented and what is being looked after?
        - Is customer value an important factor that you look at?

- How would you define the concept of customer value from your work within product development projects at the company?

    - Do you think that this definition is consistent throughout the department(s) and the company? Why/Why not?
        - Who is responsible for this?
        - Which departments or individuals are investigating and have knowledge of this?
            - Where is this information important for development?

    - How do you work with customer value throughout the development of a project?
        - Why is it important for you to take customer value into account during development?
        - Examples?

- Do you often see misinterpretations regarding customer value?
    - When does this typically happen?
    - In your opinion, what is the cause of such misinterpretations?
    - Can you think of an example of such a situation(s)?
        - Do such misinterpretations happen within or between divisions or departments?
        - How was it solved?

- How has your view of customer value changed over time?
    - What has made your view change?
        - How has this change influenced development?
        - Can you exemplify?
            - One past example with past view of customer value?
            - One recent example with current/recent view of customer value?

- Is there any difference when developing physical products or digital products in terms of customer value?
    - What do you think creates a difference between these different products?
        - Examples?
    - How does this change when developing something that has both physical and digital components?
        - Examples?

- How do you estimate value within a project? Examples?
    - Do the ways of estimating vary between digital and physical products?
        - In what ways do they vary?
        - Examples?
    - How is this estimation done for a project that has both product and service elements? (For example Husqvarna Fleet Services)

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